

HIRO FUNATSUKA

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SUMMARY

System engineer in Tokyo with 1+ year of full-time experience building applications on Amazon Web Services. Develops distributed search systems with Python; improves production reliability through retry design, load testing, and CloudWatch monitoring.

WORK EXPERIENCE

System Engineer (Full-time)

April 2025 — Present

BREXA Technology Inc.

Tokyo, Japan

- Maintain an AWS-backed bulletin-board web application (Python Flask, MySQL, OpenSearch, Lambda, SQS) used daily by front-office staff for search and content discovery.
- Improved search reliability by implementing an HTTP 429 retry policy (one retry, 500 ms wait), verified with pytest mocks and JMeter load tests.
- Monitored post-release production health via CloudWatch Logs and smoke-tested bulletin search flows; traced a recurring 500 error to a timestamp parsing defect and reported root cause to the team for remediation.
- Designed and implemented an OpenSearch sort field with `max(publish.start, last.updated)` computed in Lambda (delta-update and reindex batch), updating the Flask app to fix bulletin list ordering without a MySQL migration; presented to a non-technical client, released to staging, and awaiting client acceptance testing.
- Built React/TypeScript UI components from Figma designs for a client website project, defining shared component boundaries to reduce duplicate implementation across the team.
- Built a Voice RAG proof of concept (React, FastAPI, Twilio, OpenAI Realtime API) for call-centre automation, delivering low-latency speech-to-speech dialogue versus a Bedrock + Polly baseline.

Data Scientist (Internship)

September 2022 — March 2024

WALC Co.

Tokyo, Japan

- Led development of a Python healthcare chatbot with retrieval-augmented generation (RAG), gathering requirements from medical experts and delivering a secure user-facing application.
- Engineered an end-to-end AWS data pipeline and automated document processing workflows, improving response quality by 30% through systematic data curation.
- Built a PyTorch CNN model to classify diabetic retinopathy severity, achieving a quadratic weighted kappa of 0.88 through dataset augmentation and pre-trained architecture experimentation.

Machine Learning Engineer (Internship)

September 2022 — September 2022

Lightblue Technology Co.

Tokyo, Japan

- Built a PyTorch CNN for multi-class image classification, achieving 91.1% accuracy through architecture experimentation.

EDUCATION

Master in Science (MSci) in Computer Science, King's College London

September 2019 — August 2024

- Grade: First Class Honours
- Relevant Coursework: Software Engineering Group Project; Database Systems; Internet Systems; Object-Oriented Programming; Data Structures and Algorithms; Operating Systems and Concurrency

PROJECTS

Smart Search (gitfront.io/aws-smart-search)

November 2023 — March 2024

- Delivered a scalable vector search solution on AWS OpenSearch and SageMaker for a local council, improving search quality by 40% (NDCG) over the existing engine.
- Designed an NDCG evaluation framework with a curated test dataset and led knowledge transfer on cloud search architecture to 6 engineers in a multinational team.

Depression Monitor (gitfront.io/react-native-depression-monitor)

January 2021 — April 2021

- In a team of 9, built a React Native app with separate patient and admin interfaces for depression monitoring and simulated cognitive behavioural therapy (CBT).
- Gathered requirements from a psychology professor and implemented patient and admin dashboards with a Firebase/Firestore content management system for dynamic therapy content.

Scam Baiter (gitfront.io/rails-scam-baiter)

November 2020 — December 2020

- In a team of 5, developed a full-stack web app simulating online banking using Ruby on Rails, PostgreSQL, and Bootstrap.
- Implemented user authentication (Devise) and automated transaction history, with unit tests before Heroku deployment.

SKILLS

Programming Languages

Python, TypeScript, Java, C++

Frameworks / Libraries

React, Flask, FastAPI, PyTorch, Ruby on Rails

Cloud (AWS)

Lambda, API Gateway, EC2, DynamoDB, S3, SQS, OpenSearch, CloudWatch

Databases

MySQL, PostgreSQL, DynamoDB

Tools / Practices

pytest, JMeter, REST APIs, Docker, Git